TED(10) 4100 **Reg.No.**

Revibion2010 **Signature:**

Sixth Semester Diploma Examination in Polymer Technology

Model question Paper

PROCESS CONTROL &INSTRUMENTATION

(Maximum Marks:**100**)

Time:3Hours

PART A

(Answer all questions ,each carries **two** marks) Marks

I. 1.Define the terms calibration and fidelity.

2.Name two level measuring instruments directly.

3.What is the equivalent Fahrenheit temperature of 27^0C

4.What are the temperature control methods of calendaring process?

5.Give three advantages of pneumatic controllers. (5x2=10)

PART-B.

(Maximum Marks:30)

II. Answer any **Five** questions, each carries **Six** marks.

1.Explain the working of a mercury -in-glass thermometer with a diagram.

2.Explain the temperature measurement using pressure spring thermometer.

3.With a diagram, explain the level measurement using sight glass in an open tank.

4.Explain the working of venturimeter and state its advantages.

5.Explain the pressure measurement by piezoelectric pressure transducer.

6.What are the advantages and disadvantages of ON-FF controllers?

7.Explain the principle of hydraulic press with a diagram. (5x6=30)

PART-C

(Maximum Marks:60**)**

(Answer one full question from each module,each carries **15** marks)

Module I

III a). Explain the general characteristics of an instruments. 7

b).Explain the working of an industrial thermocouple diagram with a diagram. State its advantages. 8

OR

IV.a) Explain the method of pressure measurement using micromanometer 7

b) Explain with a neat diagram,the working of a radiation pyrometer. 8

Module II

V. a**)** Explain the working of a variable area meter with a suitable example. 7

b) Explain the principle and working of Air purge type level measuring device with sketch. 8

OR

VI.a).Explain the construction and working of float actuated liquid level Measuring instrument.State its advantages and disadvantages. 8

b)Explain the principle and working of turbine flow meter. 7

Module III

VII.a)Explain the principle and working of linear variable differential transducer. 8

b) Explain the proportional and integral control action with block diagrams. 7

OR

VIII. a)Describe the working of potentiometric pressure transducer with a diagram. State its advantages and disadvantages. 8

b) Differentiate between open loop control and closed loop control with examples. 7

Module IV

IX. a)Explain the transfer moulding process with respect to its temperature and Pressure measurements and their control. 8

b)Explain the principle and working of Hydraulic crane with a neat diagram. 7

OR

X.a) Explain the injection moulding process with a diagram.How the process parameters like temperature and pressure controlled? 8

b)Explain the different types of casings of centrifugal pumps. 7